JAMES J. CARSELLO

630-854-6430 | jamesjc4@illinois.edu | linkedin.com/in/james-carsello | https://jjcarsello.github.io/

EDUCATION

University of Illinois Urbana-Champaign (UIUC) | Grainger College of Engineering *Expected: May 2028*

Bachelor of Science in Computer Engineering with a Minor in Technology & Management through the Hoeft program

GPA: 4.00/4.00

Relevant Coursework: Digital Systems Laboratory (ECE 385), Analog Signal Processing (ECE 210), Data Structures (CS 225), Computer

Envrolled Relevant Coursework: Computer Systems Engineering (ECE 391), Applied Parallel Programming (ECE 408)

WORK EXPERIENCE

ABC Supply | Chicago, IL | Custom Development Team *May 2026 – Aug 2026*

Software Engineer Intern

- Upgraded .NET frameworks, built custom packages, and resolved bugs to support e-commerce application value streams
- Designed and published a reusable health check NuGet package to the internal registry for company-wide Azure Functions microservices, and built an ASP.NET MVC dashboard to monitor dependency connection health
- Configured hot reload for the main e-commerce site, reducing front-end developer reload times by 80%
- Held daily Agile standups, tracked tickets on Jira Kanban boards, and documented technical investigations in Confluence

PROJECTS

Space Invaders | University of Illinois Urbana-Champaign *October 2026 – Present*

- Designed a hardware-accelerated Space Invaders system-on-chip (SoC) on a Xilinx Spartan-7 FPGA. Wrote custom SystemVerilog RTL for the sprite rendering pipeline and connected it to a MicroBlaze soft-core processor over AXI4 interface.
- Built the video output path using finite-state-machine-based VGA timing logic and a VGA-to-HDMI encoder IP. Handled synchronous design across the pixel and system clock domains in Vivado.
- Wrote embedded C firmware on the MicroBlaze for real-time game-state. Brought in USB keyboard input through a USB host controller over SPI.
- Verified the RTL at unit level with SystemVerilog testbenches in simulation, then checked it on-chip at runtime with Integrated Logic Analyzer (ILA) debug cores. This covered both simulation and FPGA hardware testing.

ICompressor *Feb 2026 – Feb 2026*

- Developed a C++ file management tool with a Boost.Beast HTTP server that recommends files for deletion or compression based on last-modified date, file type, and size; demoed to 50+ attendees at Pulse Hackathon 2026
- Implemented Huffman coding for lossless file compression, achieving up to 90% file size reduction

Field Visualizer *Oct 2025 – Nov 2025*

- Built an interactive electric and magnetic field visualizer in C++ with CMake and SFML, rendering real-time field lines for user-placed charges and updating positions from net force calculations
- Implemented collision detection using charge force vectors and a user-defined coefficient of restitution

INVOLVEMENT

Liquid Rocketry at Illinois *Aug 2026 – Present*

Engine Electronics Engineer

- Develop embedded firmware using memory-mapped I/O to interface with peripherals on STM32 microcontrollers
- Integrate FreeRTOS into telemetry peripherals, enabling multithreaded task scheduling to improve real-time data collection
- Implement LwIP socket connections on STM32 microcontrollers, transitioning telemetry communication from USB to Ethernet for faster, more reliable data transfer

Disruption Lab | UIUC Gies College of Business *Jan 2026 – May 2026*

Software Engineer

- Collaborated with MariaDB Foundation engineers to develop a C++ storage engine plugin for the MariaDB RDBMS
- Supported columnar Apache Parquet files and integration with Apache Iceberg tables, reducing read times by up to 95%

SKILLS & ACHIEVEMENTS

Languages: Python, Java, C, C++, C#, Javascript, Typescript, CSS, HTML, SystemVerilog

Frameworks/Platforms: Angular, React, Jest, Stencil, Node.js, Cucumber, Dotnet, Vivado, Git, Azure Pipelines, Github Actions, Visual Studio, Visual Studio Code, Spartan 7 FPGA, Unity, SPI